ESPRIT – Engineering School of Information & Communication Technologies

4th Year Data Science – Class 4DS8

Integrated Project (PI) – Academic Year 2025/2026

ALIA

AI Avatar for Medical & Pharma Excellence

Final Report

Covering all four project phases:

- **Phase 1** – Business Understanding
- **Phase 2** – Data Acquisition & Understanding
- **Phase 3** – Modeling
- **Phase 4** – Deployment

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From field challenges to AI-driven simulation at Vital Laboratoire

Abstract

This document is the **Final Report** of the ALIA project (*AI Avatar for Medical & Pharma Excellence*), conducted by the HexaMind team as part of the Integrated Project (PI) of ESPRIT's 4th-year Data Science curriculum, in industrial partnership with **Vital Laboratoire**. It consolidates all four phases of the project lifecycle into a single, self-contained reference document.

The pharmaceutical industry requires medical representatives who can deliver scientifically rigorous, regulatory-compliant and personalised interactions with healthcare professionals under high time pressure. Existing solutions (Osler, Kognito, Quantified AI) address only fragments of this challenge. ALIA fills the gap by delivering a dual-mode AI companion: a *Training Mode* that simulates realistic medical visits with virtual doctors, and a *Commercial Mode* that acts as a real-time scientific co-pilot.

Technically, ALIA is an end-to-end multi-modal pipeline that chains **Whisper** (speech recognition), **GraphRAG** orchestrated with **LangGraph** over a **Neo4j** knowledge graph, **Llama 3.7 via Groq** (orchestration and response generation) and **ANAM AI** (real-time avatar rendering). The system achieved 93 % speech recognition accuracy, 88/100 LLM relevance score, 450 ms avatar latency and 85 % end-to-end success rate on the evaluation set. The deployment track documents **Training Mode** and **Commercial Mode** with role-based dashboards (admin, delegate, doctor); a desktop client remains on the roadmap.

Keywords: Medical AI, Pharmaceutical Training, GraphRAG, Knowledge Graph, Neo4j, LLM, Whisper, ANAM AI, Avatar, TDSP, SmPC Compliance.

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General Introduction

The pharmaceutical sector is undergoing a profound transformation. Medical representatives — the historical link between laboratories and healthcare professionals — face growing expectations: rigorous scientific mastery, strict regulatory compliance, and the ability to personalise every interaction in an environment of increasing competitive pressure. Traditional training methods, based on static presentations, occasional role-play sessions and subjective verbal feedback, can no longer keep up with these demands. They are costly, poorly standardised, and provide no objective measure of field performance.

It is within this context that the **ALIA** project was born, developed by the **HexaMind team** of ESPRIT's 4DS8 Data Science class, in partnership with **Vital Laboratoire**. ALIA — *AI Avatar for Medical & Pharma Excellence* — is an artificial intelligence companion designed exclusively for the pharmaceutical representative workflow. It combines the realism of immersive simulation, the precision of scientific knowledge retrieval, and the safety of regulatory compliance guardrails into a single, unified platform.

This final report covers the complete project lifecycle, from initial framing to deployment, across four structured phases that mirror the Microsoft Team Data Science Process (TDSP):

- **Phase 1 – Business Understanding:** problem framing, competitive analysis, solution design, business objectives, TDSP roles, SDG alignment, prototyping and user personas.
- **Phase 2 – Data Acquisition & Understanding:** data requirements, the **Vital Laboratoire** product corpus (PPT/catalogue → structured JSON), complementary Tunisian public sources (Pharmacie Centrale, Med.tn HCP), optional external medication benchmarks, exploratory analysis, risk mitigation and role-based UI design.
- **Phase 3 – Modeling:** system architecture, model selection (Whisper, LangGraph +

GraphRAG on Neo4j, Llama 3.7, ANAM AI), evaluation metrics and benchmarking.

- **Phase 4 – Deployment:** Training Mode and Commercial Mode demonstrations, role-based dashboards (administrator, delegate, doctor), and future roadmap.

Each part stands alone as a complete deliverable while also connecting to the others through a shared technical vision and a consistent set of measurable KPIs.

Part I

Business Understanding

Phase 1 — Business Understanding

This phase establishes the strategic foundations of the project: problem statement, competitive landscape, proposed solution, business and data science objectives, team organisation (TDSP), alignment with the UN Sustainable Development Goals, prototype architecture and user personas.

Chapter 1

Project Description

1.1 General Context

The ALIA project is part of the **Integrated Project (PI)** introduced by ESPRIT and conducted in partnership with **Vital Laboratoire** within the Data Science module. Its goal is to enhance the training of medical representatives and to improve the quality of their interactions with healthcare professionals through advanced Artificial Intelligence, by bridging pharmaceutical training and commercial interaction in a single, AI-driven solution tailored to the field reality of pharmaceutical companies.

1.2 Problem Statement

The training of medical representatives and the standardisation of their interactions with Healthcare Professionals (HCPs) currently suffer from several structural limitations: high cost and lack of flexibility of traditional sessions, weak standardisation of medical messaging, limited realism of role-play simulations, and absence of objective and measurable evaluation. As a result, existing approaches do not cover the entire pharmaceutical workflow, especially the combination of *training*, *performance assessment* and *commercial communication* within a unified solution.

1.3 Study of Existing Solutions

Three reference solutions were analysed to position ALIA on the competitive landscape.

Table 1.1: Comparative analysis of existing solutions.

Solution	Functionality	Limits
Osler	Uses AI to simulate patient conversations and helps train clinical communication.	Only supports clinical training; cannot simulate pharmaceutical product presentations.
Kognito	Simulates interactive conversations with AI avatars to train users in health communication.	Not designed for pharmaceutical sales or product knowledge; feedback is mostly soft-skill oriented, not performance metrics.
Quantified AI	Provides AI-powered sales role-play and coaching to improve skills and performance.	Lacks healthcare-specific or medically accurate simulations; cannot simulate real doctor or pharmacist interactions.

Market Gap

None of the existing solutions covers **simultaneously** the realism of medical simulations, the rigour of pharmaceutical product knowledge and the measurability of commercial performance. This gap is exactly the space that ALIA targets.

1.4 Proposed Solution: ALIA

ALIA is designed as a dual-mode AI companion built specifically for the pharmaceutical representative workflow.

Training Mode

Realistic role-play interactions with virtual doctors or pharmacists to practise clinical communication and objection handling. The system delivers automatic compliance and performance evaluation after each session.

Commercial Mode

A virtual presenter that delivers consistent, compliant and structured medical product presentations to HCPs, dynamically adapted to the healthcare professional's profile.

1.5 Added Value

- **Context specificity** — while platforms like Osler or Kognito focus on generic clinical or behavioural skills, ALIA is purpose-built for the Pharma / Medical Representative workflow.
- **Unified ecosystem** — combines simulation, performance analytics and commercial delivery in a single AI companion.
- **Multilingual** — tailored for diverse markets with support for **FR, EN, AR** and **ES**.

Chapter 2

Business and Data Science Objectives

The strategic ambitions of ALIA are translated into measurable data science objectives, each backed by a quantitative metric and a target KPI. This mapping is the contract that aligns the pharmaceutical ROI of Vital Laboratoire with the technical performance of the HexaMind team.

Table 2.1: Business Objectives, Data Science Objectives, Metrics and KPIs.

Business Objective	Data Science Objective	Metric	KPI (Target)
1. Improve Representative Skills	Automate performance evaluation and scientific compliance of dialogues.	% of compliant responses / Avg. performance score	$\geq$90 % compliance / +20 % score increase
2. Analyse Competitor Products	Analyse competitor products to generate personalised recommendations to delegates.	No. of comparisons generated / Prep score	100 % of competitor data analysed / High relevance
3. Standardise Medical Discourse	Verify that interactions follow scientific and commercial standards.	% of compliant dialogues / Error correction time	100 % validated messaging / Zero inconsistencies

Chapter 3

TDSP Roles

The HexaMind team adopts the Microsoft *Team Data Science Process (TDSP)* as its project-management framework. TDSP defines a structured division of responsibilities that ensures lifecycle coverage, traceability of decisions, and clean hand-offs between the business, the data science and the engineering layers.

Table 3.1: TDSP role distribution within the HexaMind team.

Role	Member	Primary Responsibility
Project Manager	**Nesrine Rouissi**	Strategic direction, stakeholder coordination, TDSP documentation, timeline and ROI definition.
Technical Lead	**Mohamed Aziz Tlili**	Technical orchestration, code validation, cross-role coordination and best-practice enforcement.
Data Scientist	**Lobna Ben Youssef**	LLM prompt engineering, RAG optimisation, data preprocessing and performance scoring models.
Data Scientist	**Bilel Brini**	Exploratory data analysis, feature engineering, persona modelling and evaluation pipelines.
Solution Architect	**Hazem Mrad**	System architecture, infrastructure setup, API design, vector and graph database integration.
Solution Architect	**Lina Haddad**	Front-end / back-end integration, deployment topology and security alignment.

Why TDSP

TDSP guarantees that every artefact produced by the team — from the business case to the deployed model — is owned by an identified role and validated by another. This is the backbone of the academic and industrial reproducibility of the ALIA project.

Chapter 4

Alignment with the Sustainable Development Goals

The ALIA project explicitly contributes to three United Nations Sustainable Development Goals (SDGs): **SDG 3** (Good Health and Well-Being), **SDG 4** (Quality Education) and **SDG 9** (Industry, Innovation and Infrastructure).

SDG 3 – Good Health and Well-Being

Target 3.c – Health Workforce.
Supporting a better-prepared healthcare workforce through improved field interactions between medical representatives and healthcare professionals, leading to safer and more effective pharmaceutical communication.

SDG 4 – Quality Education

Target 4.4 – Skills for Employment.
Strengthening medical representatives' skills for effective and professional healthcare communication, through standardised, measurable and continuous AI-driven training.

SDG 9 – Industry

Target 9.5 – Innovation, Research and Technological Capacity.
Accelerating innovation through AI-powered training and role-play for pharmaceutical field teams, contributing to the digital transformation of the pharmaceutical sector.

Chapter 5

Prototyping and User Personas

5.1 Technical Architecture

The ALIA prototype is structured around a multi-modal pipeline that connects voice input, medical knowledge retrieval and a real-time conversational avatar. The high-level architecture anticipates the detailed model choices validated in Phase 3.

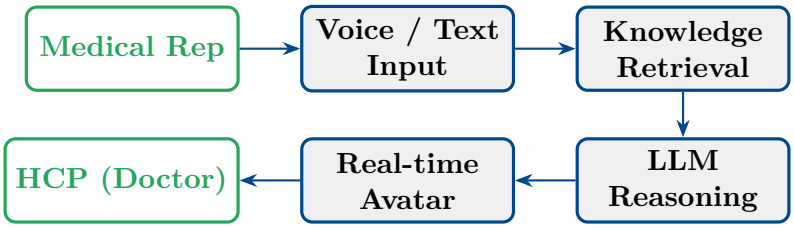


Figure 5.1: High-level prototype architecture of the ALIA dual-mode pipeline.

5.2 Dual-Mode User Interface

The prototype exposes the two operational modes through a unified interface. The mode selection screen is the unique entry point that lets the user switch between *Training Mode* and *Commercial Mode* without leaving the application, ensuring a coherent experience across simulation, evaluation and commercial delivery.

5.3 User Personas

The functional design of ALIA is anchored on two complementary personas.

5.3.1 Target Persona — Healthcare Professional (Doctor)

Healthcare Professional – Doctor

Demographic profile. Predominantly male (65 %) versus female (35 %), aged 40 to 60+ years, with significant clinical experience. Average pharmaceutical spending of approximately € 1,500 per year per practitioner.

Behavioural profile. Time-pressed professionals exposed to a high number of medical visits per week, with limited tolerance for non-scientific or non-compliant pitches. Expect rigorous, evidence-based and personalised arguments.

Implication for ALIA. The avatar must reproduce the level of demand of an experienced specialist, embed scientific rigour in every answer and adapt the discourse to the doctor's specialty and available time.

5.3.2 Target Persona — Medical Delegate

Medical Delegate – End-User of ALIA

Profile. Field professionals responsible for presenting pharmaceutical products to HCPs, building long-term scientific relationships and reporting interactions to the laboratory's CRM system.

Pain points. Limited preparation time, difficulty mastering every product's scientific dossier, exposure to hard-to-anticipate objections, and no standardised performance evaluation.

Expected value from ALIA. A realistic training environment to rehearse difficult visits, an on-demand scientific copilot during real interactions, and an automated compliance and performance evaluation after every session.

Part II

Data Acquisition & Understanding

Phase 2 — Data Acquisition & Understanding

This phase translates strategic objectives into a concrete data foundation: regulatory-safe Vital product knowledge extracted from partner artefacts, aligned with Tunisian public catalogues and HCP directories, plus a risk-mitigation register, Horizontal RAG framing and role-based UI workspaces (delegate, doctor, admin).

Chapter 6

Business and Data Science Objectives — Data Lens

This chapter does **not** redefine the business or data-science objectives: those, together with metrics and target KPIs, are stated once in Chapter 2 (Table 2.1). Here, the **same three objectives** are viewed from a *data-acquisition angle*: what technical sub-problems they imply, why ALIA's architecture is differentiated, and which datasets or feeds must exist. The first column below matches Table 2.1 verbatim.

Synthesis

The three objectives form a triangle: **Speed** (Coach), **Depth** (Strategist), **Safety** (Compliance Guard). Phase 2 verifies that the data ecosystem can sustain this triangle **given** the KPI contract in Phase 1.

Table 6.1: Phase 1 objectives (column 1) extended with a data & differentiation lens.

Business objective (Phase 1)	Technical / DS focus	Why ALIA Wins	Data Requirements
1. Improve Representative Skills	Adaptive dialogue and persona modelling for high-pressure simulation (STT, LLM, avatar loop).	**Ultra-Low Latency:** sub-300 ms perceived response where possible.	Interaction Logs, Audio Prosody, Persona Archetypes
2. Analyse Competitor Products	GraphRAG-style retrieval and cross-indexing of multi-modal clinical and market evidence.	**Deep Reasoning:** logic-based "weak spot" detection.	Med.tn, Pharmacie Centrale, PubMed, ClinicalTrials.gov, Competitor DB
3. Standardise Medical Discourse	Guardrailed LLM inference with deterministic semantic checks for SmPC-aligned messaging.	**Regulatory-First:** zero-risk, SmPC-validated output.	FDA/EMA SmPC Labels, Approved Claims, CRM History

Chapter 7

Data Requirements and Data Understanding

7.1 Data Requirements

ALIA combines **authoritative partner product knowledge** with **Tunisian public reference data** and, where useful, **external scientific corpora** for competitive and clinical context:

- **Vital product corpus (primary):** structured records for Vital Laboratoire commercial ranges, derived from internal training and catalogue artefacts (see Section 7.2).
- **Local market calibration:** Pharmacie Centrale de Tunisie (PCT) — national authorised human medicines list; Med.tn — HCP directory and practice metadata.
- **Behavioural / simulation data:** interaction logs and audio prosody samples for Training Mode coaching (public or project-generated, depending on release).
- **Broader clinical and regulatory context:** PubMed, ClinicalTrials.gov, and FDA/EMA SmPC-style references for GraphRAG guardrails and competitor reasoning.

7.2 Vital product corpus — from PPT and catalogues to JSON

7.2.1 Raw sources and repository layout

Partner material that is **not suitable for public redistribution** (marketing PPTX, slides, scanned or exported catalogue sheets) is kept under the project folder `resources/catalogue/`. This directory holds the **raw acquisition layer**: original PowerPoint decks, PDF extracts, and any intermediate tables produced during cataloguing. Versioning follows the partner hand-offs (file dates and deck titles are preserved in folder names where applicable).

7.2.2 Extraction and structuring workflow

The team normalised this heterogeneous material into a single machine-readable artefact:

1. **Slide and document ingestion** — each PPTX is processed slide-by-slide; text regions, bullet lists and tabular blocks are collected. Catalogue pages (PDF or spreadsheet exports) are aligned with the corresponding commercial ranges (e.g. hygiene, paediatrics, OTC families) so that naming stays consistent with field use.

2. **Field mapping** — free text is mapped to a fixed JSON schema: commercial name, brand, galenic form, volume/weight, category/gamme, indications, target population, posology instructions, route of administration, and structured `composition` entries (ingredient + concentration). Optional `metadata` blocks record provenance (e.g. `source_slide` index) for audit and manual correction.

3. **Cleaning and validation** — trimming and Unicode normalisation; removal of duplicate bullets; harmonisation of units (%, mg/ml, alcohol proof labels); split of multi-ingredient lines into separate composition objects; French medical wording kept as provided by Vital for delegate-facing consistency.

4. **Consolidation** — all validated product objects are merged into one array under a top-level `products` key for downstream ingestion.

7.2.3 Deliverable file and usage

The consolidated dataset is stored at `data/data_vital(1).json`. At report time it contains **145** distinct product records. This file is the **reference Vital layer** for Neo4j graph seeding (see `scripts/ingest_to_neo4j.py`), Supabase product seeding, and in-app "Vital catalogue" views. It supersedes earlier placeholder narratives based solely on generic open medication tables: those sources may still inform competitor enrichment, but they are no longer described as the primary scientific ground truth for Vital brands.

Traceability

Every record can be traced back to a specific slide or catalogue excerpt via `metadata`, supporting partner review and SmPC-aligned corrections without re-running the full pipeline.

7.3 Data Understanding — Pharmacie Centrale Dataset

- **Source:** Pharmacie Centrale de Tunisie — national public catalog.
- **Size:** 577 observations across 13 variables.
- **Key features:** Commercial Name, Dosage, Pharmaceutical Form, Therapeutic Class, Sub-class, Manufacturer.

This catalog is the **local calibration reference**: every product from the broader Medicine Dataset is cross-checked against Pharmacie Centrale to confirm local availability, official dosage and reimbursement status — a critical verification since a Tunisian representative cannot promote a product not registered locally.

7.4 Data Understanding — Med.tn HCP Dataset

- **Source:** Med.tn — Tunisia's leading healthcare professional directory.
- **Size:** 1,440 observations across 13 variables.
- **Key features:** Doctor Name, Specialty, City, Address, Phone Numbers, CNAM Coverage, and Weekly Schedule (Monday–Saturday).

The dataset covers the principal Tunisian governorates (concentration in Greater Tunis, followed by Sfax and Sousse). The weekly schedule fields enable the *visit-planning* feature of the Strategist module, a competitive differentiator versus Osler and Kognito.

7.5 Ingestion tooling and graph preparation pipeline

This subsection records **implementation-level** details that apply across sources complementing the Vital catalogue (Section 7.2):

- **Web scraping** — public HTML catalogues and listings (notably `med.tn`, Pharmacie Centrale portals, and selected FDA / SmPC pages) are collected using

BeautifulSoup in Python, with retries and parsing rules aligned to each site's markup.

- **OCR** — when native text is unavailable, **optical character recognition** is applied to regulatory leaflets, product packaging images, or legacy PDF scans before normalisation.
- **CRM (VITAL)** — internal delegate–HCP interaction history supplied by the partner is merged only after compliance approval; it uses the same cleaning and identifier conventions as the public feeds.

Raw heterogeneous records then pass through the six-step pipeline in Table 7.1 before exposure to the LLM and GraphRAG layers.

Table 7.1: Data preparation pipeline (all streams).

Step	Action	Objective
Cleaning	Remove unnecessary values, fix formats, remove duplicates	Clean data
Normalisation	Standardise names, INN (DCI), classes, laboratories	Avoid inconsistencies
Merging	Combine Vital JSON + `med.tn` + Pharmacie Centrale (PCT)	Complete dataset
Enrichment	Add additional information from external sources	Enriched data
Structuring	JSON format with unique IDs	Organised data
Graph Modelling	Create relationships using **Neo4j**	Link drugs, classes, laboratories

Output

A unified, deduplicated and graph-ready knowledge base linking *molecules* → *therapeutic classes* → *products* → *laboratories*, ready to feed the GraphRAG layer.

Chapter 8

Reflection on Data Understanding

The exploration revealed three structural risks, each addressed by a concrete engineering response.

Table 8.1: Data Reflection and Risk Mitigation Register.

Challenge	Solution
Heterogeneous partner formats (PPTX, PDF) vs stable graph schema	**Catalogue → JSON contract** documented in Section 7.2, with provenance metadata and a single consolidated file (`data/data_vital(1).json`) ingested by the same pipelines used for public sources.
Multi-source and multi-format data increases integration complexity	A **unified preprocessing pipeline** standardises and connects all data sources into a single warehouse with controlled vocabularies.
Training data contains disease and regional bias underrepresenting the Tunisian context	Apply **weighted sampling** on under-represented diseases and integrate Pharmacie Centrale to inject local prevalence priors.

Key Take-away

The data understanding phase ends with a **quantified risk register** and a set of mitigation pipelines that make the project resilient to data shifts and to the late arrival of production data.

Chapter 9

Modeling Architecture and UI Mock-ups

9.1 Global System Architecture — Horizontal RAG

ALIA adopts a **Horizontal Retrieval-Augmented Generation (RAG)** architecture. Unlike classical vertical RAG (single corpus), the Horizontal RAG retrieves *simultaneously and in parallel* from: a vector store for scientific literature, a Neo4j knowledge graph for relational queries, a regulatory store for SmPC labels, and a CRM store for historical interactions. Retrieved contexts are merged by an orchestration layer implementing the cyclical logic *fetch* → *verify* → *re-run if non-compliant*.

9.2 Modeling Strategy and Technology Choices

- **Whisper:** speech-to-text (STT) for robust transcription of delegate voice input (including French and Arabic field conditions).
- **LangGraph:** workflow library used to implement GraphRAG—orchestrating retrieval, reasoning steps and compliance-oriented cycles over the knowledge graph.
- **Neo4j:** graph database for **GraphRAG schematisation**—entities and relationships (products, molecules, indications, HCP context) are modelled in the graph and queried for multi-hop retrieval; historical interaction data can extend the same schema over time.
- **Llama 3.7:** large language model used as the **orchestrator** and for **natural-language generation**, combining retrieved graph context with dialogue policy and guardrails.

- **ANAM AI: avatar rendering layer**—real-time visual avatar, voice output and lip-sync, replacing earlier speech-to-speech prototypes (e.g. PersonaPlex) with a decomposed STT → LLM → avatar pipeline.

9.3 Proposed UI Mock-ups

The React application (`src/`) does **not** rely on a single "mode selection" page. After authentication, each user is routed to a **dedicated workspace** that matches their profile role (`delegate`, `doctor`, or `admin`): see `SignInPage.jsx` and `ProtectedRoute.jsx`, with top-level routes in `App.jsx` (`/delegate`, `/doctor`, `/admin`). The following surfaces operationalise the data and modeling layers:

1. **Medical delegate workspace** (`/delegate`) — shell dashboard with role-specific navigation (overview KPIs, training hub, simulations, learning path, product library, HCP profiles, AI copilot, calendar, messaging). Delegates use **training-oriented** avatars and simulation flows to rehearse visits, review results, and prepare for the field.

2. **Doctor / HCP workspace** (`/doctor`) — separate shell for healthcare professionals interacting with **commercial** delegate avatars: receive structured pitches, challenge claims, and ask product-level questions in a meeting or pitch context (e.g. meeting and pitch views tied to avatar sessions), with access to product information and delegate profiles as implemented in the doctor routes.

3. **Admin workspace** (`/admin`) — intermediate **operational** role: consolidated overview, coordination and dispatch-style views, training-session oversight where enabled, and other admin-only modules (calendar, node management, feature-flagged review screens) for governance between delegates and HCP-facing experiences.

4. **Role-specific overview dashboards** — each workspace opens on an `OverviewView` tuned to the role (metrics, quick actions, activity feeds) so performance and scientific activity remain visible without mixing training and commercial contexts in one generic home screen.

5. **Live session UI** — shared patterns for real-time dialogue: voice/transcript presentation, animated avatar output (ANAM), and—where configured—compliance or guardrail feedback during simulations (delegate) or commercial interactions (doctor).

6. **Cross-cutting tools** — product catalogue views, AI copilot / knowledge queries, calendar and messaging, and unified profile settings, reachable from the appropriate role shell so each persona keeps a consistent navigation model.

Part III

Modeling

Phase 3 — Modeling

This phase converts the validated data foundation into a working intelligent system. It details the end-to-end pipeline, justifies every model choice, reports evaluation metrics and documents the benchmarking that drove three major component upgrades.

Data sources, Vital JSON extraction, and the six-step preparation pipeline are documented once in Phase 2 (Chapter 7, Sections 7.2–7.5) to avoid duplication; modeling below assumes that graph-ready knowledge base as input.

Chapter 10

System Architecture and Model Selection

10.1 System Architecture and Global Pipeline

ALIA is built as an end-to-end multi-modal pipeline chaining transcription, graph-backed retrieval, language orchestration and avatar rendering to support a real-time conversation between a medical representative and the virtual doctor.

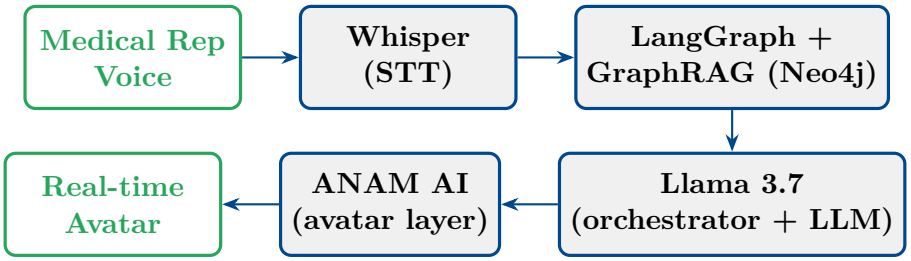


Figure 10.1: ALIA global pipeline: voice → text → LangGraph GraphRAG on Neo4j → Llama 3.7 → ANAM avatar rendering.

The pipeline follows four steps:

1. **Voice** → **Text** via Whisper (STT).
2. **Text** → **Graph context** via **LangGraph**-orchestrated GraphRAG over the **Neo4j** schema (multi-hop retrieval and compliance-oriented routing).
3. **Context** → **Response** via **Llama 3.7** (orchestration and generation), with inference served through the **Groq API**.
4. **Response** → **Avatar** via **ANAM AI** as the rendering layer (voice synthesis, lip-sync and real-time character output).

10.2 Justification of Model Choices

Whisper — Speech-to-Text

- Open-source and free (OpenAI).
- Robust to accents and background noise.
- Higher accuracy than Deepgram on French and Arabic samples.
- Easy integration into any Python pipeline.

LangGraph + GraphRAG on Neo4j

- **Neo4j** stores the domain schema: drugs, classes, laboratories, indications and links to Vital / PCT / HCP entities for multi-hop queries.
- **LangGraph** encodes the GraphRAG workflow: retrieval steps, merges with vector or regulatory context where needed, and cyclical *fetch* → *verify* logic for compliance.
- Contextual retrieval beyond flat chunk RAG; answers paths such as molecule → family → competitor.
- Same stack scales when CRM and interaction history are added to the graph.

Llama 3.7 — Orchestration & Generation (via Groq API)

- Latest Llama 3.x generation (Meta AI) used for both workflow decisions and natural-language replies to the user.
- Groq API provides low-latency inference (<1 s typical) suitable for live simulation.
- Strong instruction following for medical phrasing, persona and SmPC-aware prompts.
- Suited for long, precise answers grounded in retrieved graph context.

ANAM AI — Avatar Rendering Layer

- Real-time avatar: voice output, lip-sync and presentation—the **display** layer for the pipeline (replaces end-to-end speech-to-speech approaches).
- Decoupled from STT/LLM: Whisper and Llama 3.7 feed text or structured cues into ANAM for consistent delegate experience.
- Low end-to-end latency, essential for live training and commercial modes.
- Single integration point for immersive doctor / HCP personas.

Chapter 11

Evaluation

The complete pipeline was benchmarked end-to-end on a held-out evaluation set. The four headline metrics all pass their respective acceptance criteria.

Table 11.1: Headline evaluation metrics of the ALIA pipeline.

Metric	Value	Status
STT (Speech-to-Text) Accuracy — WER	93 %	Pass
LLM Response Relevance	88/100	Pass
Avatar Latency	450 ms	Pass
End-to-End Success Rate	85 %	Pass

Interpretation

The combination of high transcription accuracy, relevant responses, low avatar latency and a strong end-to-end success rate **validates the modeling choices** made in Chapter 10 and confirms that the system is suitable for production use.

Chapter 12

Benchmarking

The current pipeline is the result of a deliberate experimentation phase. Three component families were upgraded after empirical comparisons and field-relevant criteria (accuracy, language coverage, API maturity, cost, and hallucination risk).

Table 12.1: Evolution of technical choices after experimentation.

Component	Before	After	Summary rationale
Speech-to-Text (STT)	Deepgram	**Whisper**	Accuracy, Arabic support, and lexical reliability over raw latency (see below).
Avatar / Simulation	HeyGen (sandbox / tests)	**ANAM AI**	API flexibility, pricing, and real-time immersion versus HeyGen limits (see below).
LLM (orchestration + generation)	Llama 7B Instruct	**Llama 3.7**	Stronger reasoning, fewer hallucinations, better medical grounding (see below).

12.0.1 Prior stack: limitations observed in the project

The following points summarise **why** the earlier tools were set aside; they complement the summary column in Table 12.1.

- **Deepgram (STT)** — offered very low streaming latency, but in our tests **word-error and robustness** were insufficient for pharmaceutical dialogue. **Arabic**

was not a practical option for delegate-facing use, and the model could **invent or substitute words** (hallucinated tokens) in a way that is unacceptable for compliance-sensitive transcripts. Whisper trades a small latency cost for materially better French/Arabic behaviour and more stable decoding.

- **HeyGen (live avatar)** — the **API and sandbox workflow** were too restrictive for iterative product and test builds; integration and iteration were slower than with ANAM for our use case. **Commercial pricing** was also a constraint relative to the value delivered. ANAM provided a better fit for real-time simulation, lip-sync quality, and go-to-market flexibility.

- **Llama 7B Instruct (orchestrator + generation)** — was lightweight and fast, but **too weak for long medical answers** and **prone to hallucination** when combining persona, product facts, and retrieved context. Moving to **Llama 3.7** materially improved instruction following, factual grounding with GraphRAG context, and safety of phrasing under SmPC-style prompts.

Lesson Learned

A modeling phase is not a linear "pick the best paper" exercise. Every retained component had to win on **measurable accuracy**, **language and regulatory fit**, and **qualitative user experience**—not on a single headline metric such as STT latency alone.

Part IV

Deployment

Phase 4 — Deployment

This phase operationalises the validated pipeline: **Training Mode** and **Commercial Mode** in the product, and **role-based dashboards** for administrator, medical delegate, and doctor personas—with a forward-looking roadmap.

Chapter 13

Deployment Progress

13.1 Training Mode

The first end-to-end demonstration of ALIA's **Training Mode** was successfully completed and shared with the jury. The demo showcases the full pipeline in a live session: the medical representative speaks to the system (voice in), Whisper transcribes in real time, LangGraph-orchestrated GraphRAG on Neo4j retrieves the relevant clinical context, Llama 3.7 generates a doctor-persona response, and ANAM AI renders it through the animated avatar (voice out, with lip-sync). The session ends with an automatic compliance and performance evaluation report.

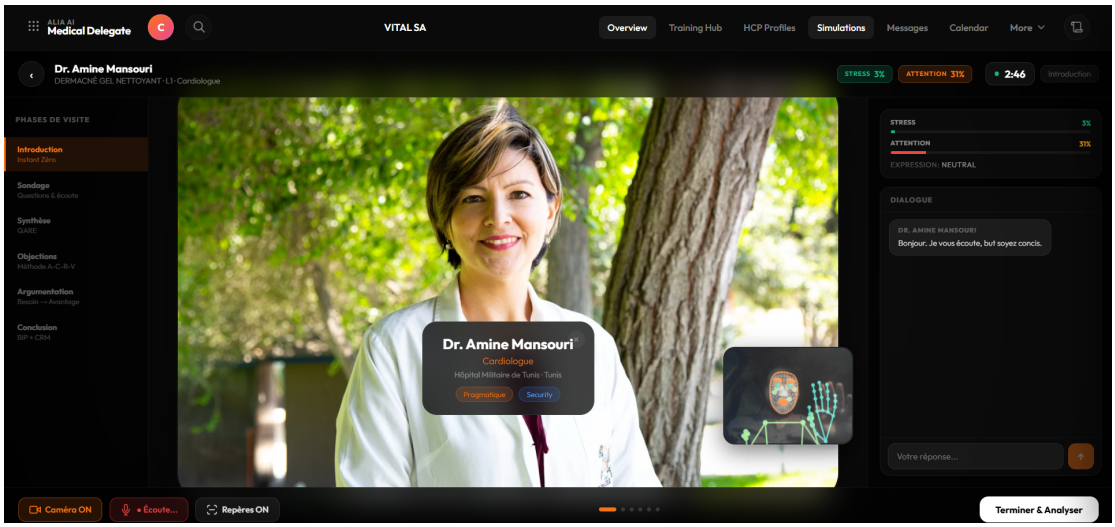


Figure 13.1: Training Mode: simulation workspace (coach session, avatar stage, and session flow).

13.2 Commercial Mode

Commercial Mode supports the delegate during real HCP interactions: scientific copiloting, product-grounded answers, and traceable visit context—complementing the rehearsal-focused Training Mode with a field-oriented workflow.

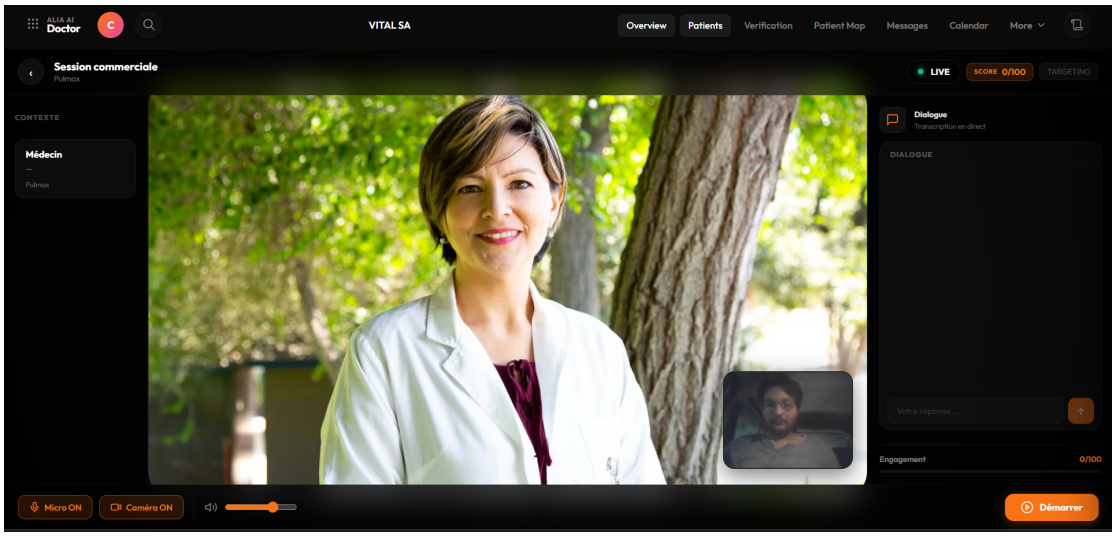


Figure 13.2: Commercial Mode: delegate-facing workflow for live visits and copilot use.

13.3 Role-based dashboards

The product exposes three primary dashboard experiences aligned with Vital's org chart: **administrator** (oversight, configuration, cohort analytics), **medical delegate** (simulations, sessions, performance), and **doctor** (meeting and clinician-oriented views).

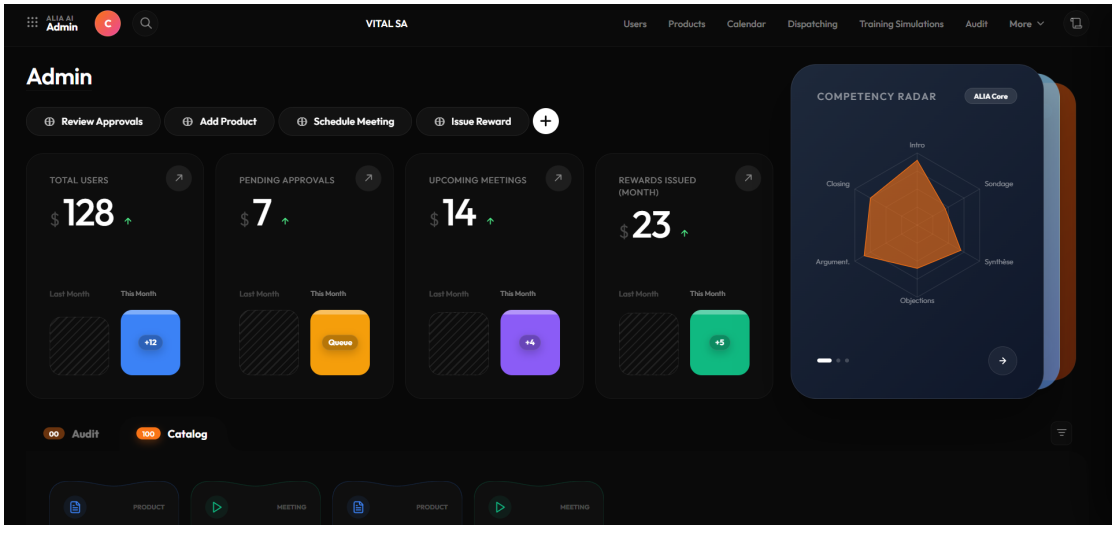


Figure 13.3: Administrator dashboard: oversight, users, and global analytics.

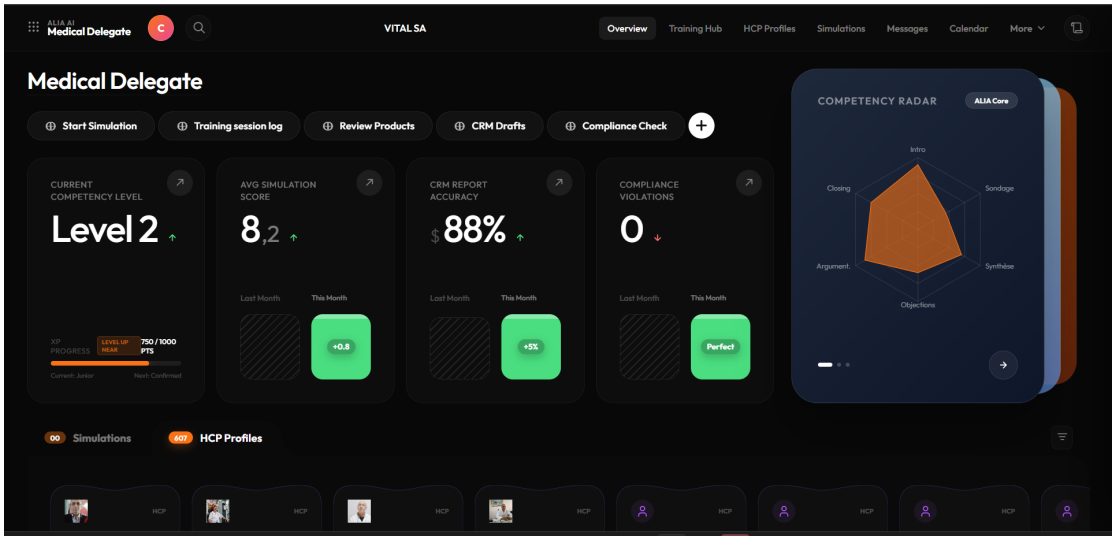


Figure 13.4: Medical delegate dashboard: training and commercial workflows, history, and metrics.

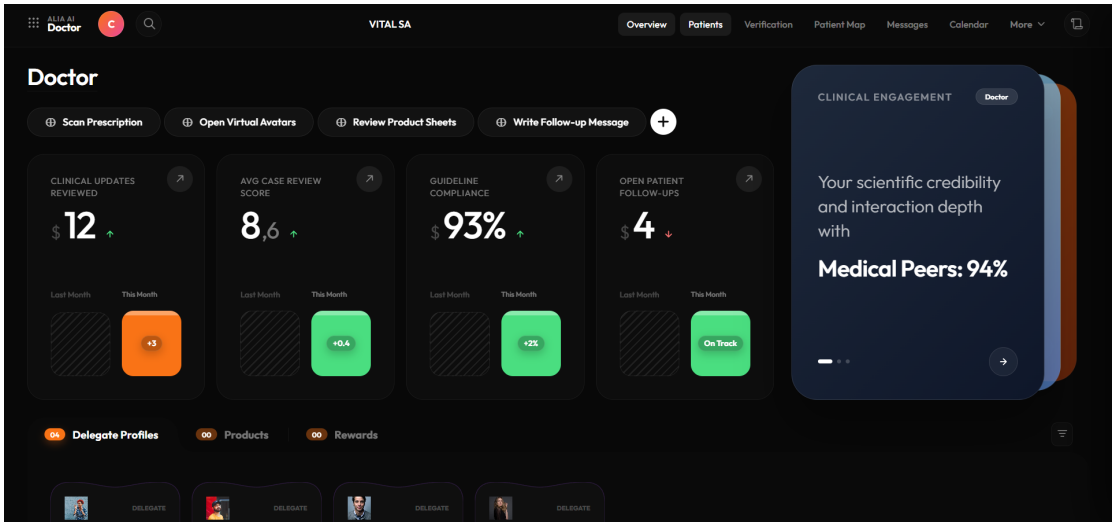


Figure 13.5: Doctor dashboard: sessions, meetings, and clinician-oriented views.

Chapter 14

Perspectives and Roadmap

The deployment of the current prototype is only the first milestone. Three strategic directions have been identified for the next development horizon:

1. **Desktop Application** — develop a full-screen desktop client optimised for in-office training sessions, with richer data visualisation and multi-monitor support.

2. **Emotion-Aware Coaching** — enable the avatar to analyse the medical delegate's facial expressions and voice prosody in real time, adapting the difficulty and feedback of the session to the representative's emotional state.

3. **CRM Integration** — finalise the avatar's CRM component to ensure 100 % traceability of every interaction: visit summary, compliance score, detected weak spots, and follow-up recommendations, all pushed automatically to the Vital Laboratoire CRM.

Roadmap Summary

Short term	Desktop application	Q3 2026
Medium term	Emotion-aware adaptive coaching	Q4 2026
Long term	Full CRM integration & multi-lab scaling	2027

General Conclusion

The ALIA project demonstrates that it is possible to build, validate and deploy a production-grade AI system for the pharmaceutical representative workflow within a single academic year, following a rigorous four-phase methodology.

Phase 1 established the strategic foundations: a clearly identified problem, a benchmarked market gap, a dual-mode solution, measurable KPIs aligned with Vital Laboratoire's ROI expectations, a complete TDSP team, and alignment with SDGs 3, 4 and 9.

Phase 2 validated the data foundation: the Vital product corpus (145 structured records in `data/data_vital(1).json`, sourced from `resources/catalogue/`), 577 Pharmacie Centrale entries, 1,440 Med.tn HCP profiles, plus optional external benchmarks for competitor analysis; a risk mitigation register; a Horizontal RAG architecture; and role-based UI workspaces in `src/`.

Phase 3 delivered the intelligent system: Whisper (STT), LangGraph-orchestrated GraphRAG on Neo4j, Llama 3.7 (orchestrator + LLM via Groq), and ANAM AI as the avatar layer—validated by 93 % STT accuracy, 88/100 LLM relevance, 450 ms avatar latency and 85 % end-to-end success, after benchmarking that replaced Deepgram with Whisper, HeyGen with ANAM AI, Llama 7B Instruct with Llama 3.7, and early speech-to-speech prototypes with the decomposed stack.

Phase 4 operationalised the vision: Training Mode and Commercial Mode demonstrations, role-specific dashboards (admin, delegate, doctor), and a clear roadmap toward desktop deployment, emotion-aware coaching and CRM integration.

ALIA is not simply a proof of concept. It is a **scalable, measurable and domain-specific AI platform** ready for industrial deployment and large-scale adoption in the pharmaceutical sector. The HexaMind team closes this project confident that ALIA meets both the academic standards of the PFE and the industrial expectations of Vital Laboratoire — and that it lays the groundwork for the next generation of pharmaceutical AI tools.